

Exam Date & Time: 02-Jan-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech VII Semester - Make up Examination, Jan-2025

SYSTEM ON CHIP DESIGN [ECE-4087]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A processor die (1.4cm X 1.4cm) will be produced for five years. Over this period, defect densities are expected to drop linearly from 1.5 defects/cm² to 0.8 defects/cm². The cost of 6-inch wafer production will fall linearly from \$5,000 to \$3,500, and the cost of 8-inch wafer production will fall linearly from \$10,000 to \$6,500. Assume production of good devices is constant in each year. Which production process should be chosen? (5)
 - A)
 - B)
 - C)
- 2) Design a Datapath for dividing two numbers input by the user, using adder/subtraction, comparator units. Identify the high-level state diagram for the machine. (5)
 - A)
 - B)
 - C)
- 3) (4)
 - A)

A four-segment pipeline implements a function and has the following delays for each segment($b=0.2$):

Segment#	Maximum delay*
1	1.7ns
2	1.5ns
3	1.9ns
4	1.4ns

*Excludes clock overhead of 0.2ns

- What is the cycle time that maximizes performance without allocating multiple cycles to a segment?
- What is the total time to execute the function (through all stages)
- What is the cycle time that maximizes performance if each segment can be partitioned into sub systems?

- Use K-mean algorithm to cluster the following 7 examples into 2 clusters: $A_1 = (185, 72)$, $A_2 = (170, 56)$, $A_3 = (168, 60)$, $A_4 = (179, 68)$, $A_5 = (182, 72)$, $A_6 = (188, 77)$, $A_7 = (180, 71)$. (4)
 - Explain the building blocks of NoC with a neat diagram. (2)
- 4) What is arbitration? Explain the difference between priority based and TDMA based arbitration protocols. (5)
- - What is the significance of routing in NoC? Explain anyone routing algorithms of NoC. (3)
 - Differentiate between 2-D mesh and torus topologies of NoC. (2)
- 5) Find the minimum number of Test vectors to detect stuck at faults at all the given points (a, b, c, d, f, g, h) of the circuit shown in Fig.5a

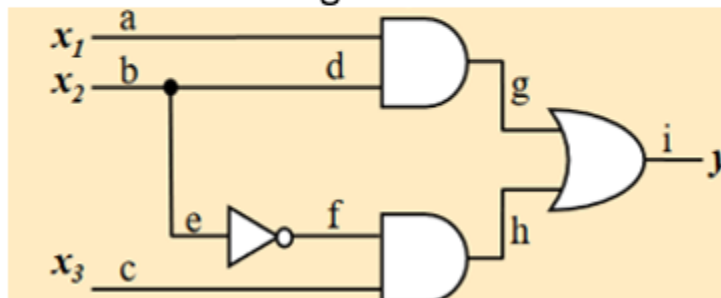


Fig.5a

- What is clock tree synthesis? Draw the structures of different types of clock trees. (3)
- Classify the IP's based on size and explain them. (2)

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